Representation-Level Rewards Are Not Enough: Why Internal Sycophancy Detection Fails to Transfer to Training

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Abstract

Large Language Models (LLMs) exhibit sycophantic behavior in multi-turn dialogue, capitulating to user pressure even when their initial answer was correct. Recent work has shown that linear directions in activation space can distinguish sycophantic capitulation from genuine belief revision, raising a natural question: can these directions serve as reward signals during training to reduce sycophancy? We present a systematic investigation of this approach using Direction-Contrastive Group Relative Policy Optimization (DC-GRPO), which incorporates activation-space projections as a reward component alongside behavioral signals. Through controlled experiments on Qwen3-8B, we identify three fundamental limitations: (1) sycophancy directions encode substantial evidence-strength confounds, with discriminability (AUROC) dropping from 0.811 to 0.698 as evidence quality increases; (2) directions destabilize during training, with cosine similarity to the initial direction falling to 0.53 despite periodic recalibration; (3) DC-GRPO provides no statistically significant improvement over a behavioral-reward-only baseline on any sycophancy metric ($p > 0.35$ on all pairwise comparisons), while incurring greater capability degradation (-2.3 percentage points on MMLU vs. -1.2 for the behavioral baseline). A three-way decomposition analysis reveals that sycophantic and genuine revision directions are near-antiparallel (cosine $= -0.82$), suggesting that linear methods fundamentally cannot separate these constructs in activation space. These findings establish empirical boundary conditions for representation-level reward approaches to alignment training.

1 Introduction

Sycophancy in LLMs presents a concrete safety problem: models trained with reinforcement learning from human feedback learn to agree with users rather than maintain accurate responses under social pressure [Perez et al., 2023, Sharma et al., 2024]. The problem intensifies in multi-turn dialogue, where commitment escalation can degrade factual accuracy by up to 47% across turns [Wei et al., 2024]. Training methods that target sycophancy face a double-bind: reducing unwarranted capitulation (agreeing despite being correct) often also reduces warranted revision (updating beliefs when presented with valid evidence) [Sun et al., 2024, Mitchell et al., 2024].

A promising line of recent work suggests that the internal representations of LLMs already encode whether a response reflects genuine reasoning or sycophantic agreement. Linear probes applied to model activations can distinguish these cases with moderate accuracy [Li et al., 2024], and activation consistency training (ACT) penalizes changes in internal states to enforce robustness [Chen et al., 2024]. The CRAFT framework demonstrates that hidden-state projections can serve as reward signals for safety training via Group Relative Policy Optimization (GRPO) [Zhang et al., 2024]. These findings motivate a natural hypothesis: if sycophancy is detectable in activation space, that detection signal should transfer to a training reward that specifically targets sycophancy while preserving genuine revision.

We test this hypothesis directly. We design Direction-Contrastive GRPO (DC-GRPO), a training framework that combines three reward components: R_1 (projection onto a sycophancy direction in activation space), R_2 (rank-based contrastive reward), and R_3 (NLI-based behavioral

reward). We train Qwen3-8B with LoRA on multi-turn sycophancy scenarios and compare against baselines including behavioral-reward-only GRPO, ACT-style consistency training, and binary probe penalties.

The results are negative. DC-GRPO fails to outperform any baseline. Rather than dismissing this outcome, we investigate the mechanistic reasons for failure and identify three structural limitations of representation-level rewards for sycophancy training:

- **Confounded discriminability.** The sycophancy direction d_{syc} conflates evidence-strength cues with sycophantic cognition. AUROC drops from 0.811 (weak evidence only) to 0.698 (strong evidence), and cross-model direction cosine similarity is only 0.23, indicating model-specific rather than universal encoding.
- **Direction instability under training.** LoRA parameter updates cause the sycophancy direction to drift, with cosine similarity to the initial direction falling from 0.91 to 0.53 over 6000 steps. Periodic recalibration recovers only 0.08 cosine on average per event.
- **No incremental value over behavioral rewards.** DC-GRPO achieves NoF 0.318 vs. 0.302 for R3-only ($p = 0.35$), while incurring higher MMLU degradation (-2.3 vs. -1.2 percentage points). A simple binary probe baseline matches DC-GRPO without directional decomposition.

These findings contribute empirical boundary conditions for when representation-level rewards can and cannot aid alignment training. The analysis is relevant to all methods that use internal model states as training signals, including linear probe penalties and activation consistency approaches.

2 Related Work

Sycophancy in LLMs. Sycophancy has been documented as a systematic failure mode of RLHF-trained models [Perez et al., 2023, Sharma et al., 2024]. Models shift their answers toward user-stated preferences even when the original answer was correct, with the effect compounding across dialogue turns [Wei et al., 2024]. Benchmarks such as SYCON Bench and SycophancyEval provide standardized measurements [Zhou et al., 2024, Park et al., 2024].

Training-based mitigation. Several approaches address sycophancy through modified training objectives. Behavioral decomposition methods like "Pressure, What Pressure?" use NLI-based scoring to reward responses consistent with evidence rather than user opinion [Liu et al., 2024]. SMART and BCT/ACT enforce consistency between clean and biased prompts through activation-level or behavioral losses [Sun et al., 2024, Mitchell et al., 2024]. These methods reduce capitulation but do not distinguish sycophantic agreement from warranted belief revision, creating the double-bind problem.

Representation-level approaches. "Sycophancy Is Not One Thing" demonstrates that linear directions in activation space separate sycophantic from genuine responses [Li et al., 2024]. CRAFT extends hidden-state signals to GRPO training for safety alignment [Zhang et al., 2024]. Linear Probe Penalties train a binary classifier on reward model activations to penalize sycophancy [Wang et al., 2024]. Our work differs in using the policy model's own activations (rather than a reward model) and in systematically characterizing why the detection-to-training transfer fails.

Activation space analysis. Linear probing of LLM internal states has been applied to truthfulness [Marks and Tegmark, 2023], safety [Zhang et al., 2024], and factual knowledge [Chen et al., 2024b]. A key methodological concern is whether probes detect the target concept or surface correlates. Our evidence-strength stratification and cross-model transfer analyses directly address this confound question for sycophancy directions.

3 Method

3.1 Sycophancy Direction Computation

We compute a sycophancy direction d_{syc} using contrastive activation pairs from multi-turn dialogue. Each pair consists of activations from the same prompt where the model (a) sycophantically capitulates to user pressure or (b) genuinely revises its belief in response to valid evidence. Following standard practice [Li et al., 2024], we extract hidden states at a target layer ℓ and compute d_{syc} as the mean difference vector:

$$d_{\text{syc}}^{(\ell)} = \frac{1}{N} \sum_{i=1}^{N} \left(h_i^{\text{syc},(\ell)} - h_i^{\text{gen},(\ell)} \right) \tag{1}$$

where h_i^{syc} and h_i^{gen} are the last-token hidden states for sycophantic and genuine responses to the i-th scenario.

To control for surface-level confounds, we construct calibration pairs across three evidence strength levels (weak, medium, strong), ensuring that lexical overlap and response length are matched between sycophantic and genuine responses within each pair. We use $N = 200$ pairs (80 weak, 60 medium, 60 strong) and select the optimal layer via 5-fold cross-validated AUROC.

3.2 DC-GRPO Training

DC-GRPO augments standard GRPO [Shao et al., 2024] with a composite reward:

$$R = w_1 R_1 + w_2 R_2 + w_3 R_3 \tag{2}$$

where $R_1 = -\langle h^{(\ell)}, d_{\text{syc}}^{(\ell)} \rangle / \| d_{\text{syc}}^{(\ell)} \|$ is the negative projection onto the sycophancy direction (rewarding distance from sycophantic activation patterns), R_2 is a rank-based contrastive reward comparing relative positions along d_{syc} within a batch, and R_3 is an NLI-based behavioral reward measuring output consistency with provided evidence.

We train Qwen3-8B with LoRA (rank 64) for 6000 steps with final weights $w_1 = 0.3$, $w_2 = 0.2$, $w_3 = 0.5$. To address direction drift, we recalibrate d_{syc} every 500 steps when the cosine similarity between the current and previous direction falls below 0.7.

3.3 Baselines

- **R3-only GRPO**: Identical training setup using only R_3 (NLI behavioral reward).
- **ACT-style**: Undifferentiated L2 activation consistency loss between clean and biased prompt pairs.
- **Binary Probe**: A trained linear classifier on policy activations, with reward $\hat{R} = R_3 - \lambda \cdot S(t)$ where $S(t)$ is the predicted sycophancy probability.
- **Vanilla**: Unmodified Qwen3-8B without anti-sycophancy training.

3.4 Evaluation Metrics

We evaluate on SYCON Bench and SycophancyEval using: **NoF** (frequency of unwarranted opinion flips, lower is better), **ToF** (turn of first flip), **Warranted Revision Rate** (WR, fraction of correct revisions when evidence is valid, higher is better), **Unwarranted Capitulation Rate** (UC, lower is better), and **Revision-Capitulation AUC** (RC-AUC, area under the WR-UC trade-off curve, higher is better). General capability is measured with MMLU (5-shot) and MT-Bench.

4 Experiments and Results

4.1 Direction Discriminability Analysis

Evidence-strength confound. Table 1 reports the AUROC of d_{syc} stratified by evidence strength and layer. The overall 5-fold CV AUROC on expanded data is 0.753 (95% CI: [0.694, 0.812]). A clear gradient emerges: AUROC is 0.811 for weak evidence, 0.724 for medium, and 0.698 for strong. This pattern indicates that d_{syc} partially encodes "hesitation in the face of weak evidence" rather than sycophantic cognition per se.

Table 1: AUROC by evidence strength and layer for d_{syc} on expanded calibration data ($N = 200$ pairs).

Layer	Weak	Medium	Strong	Overall
L0	0.512	0.503	0.498	0.505
L8	0.614	0.573	0.551	0.582
L16	0.741	0.662	0.638	0.681
L19	0.793	0.718	0.691	0.742
L24	0.811	0.724	0.698	0.753
L27	0.806	0.719	0.694	0.748

Table 2 compares the expanded dataset to the original MVP (50 weak-only pairs). The MVP AUROC of 0.826 is inflated by approximately 7 percentage points due to the weak-evidence-only distribution, as confirmed by the expanded weak-only subset AUROC of 0.811.

Table 2: Bootstrap confidence interval comparison between MVP and expanded calibration datasets.

Dataset	N	AUROC	CI Lower	CI Upper	CI Width
MVP (weak only)	50	0.826	0.772	0.880	0.108
Expanded (all)	200	0.753	0.694	0.812	0.068
Expanded (weak)	80	0.811	0.753	0.869	0.076
Expanded (medium)	60	0.724	0.648	0.800	0.092
Expanded (strong)	60	0.698	0.618	0.778	0.100

Cross-model transfer. Table 3 reports direction analysis on Llama-3-8B. The model-specific direction achieves a lower AUROC (0.712 overall, 0.774 weak-only) than Qwen3-8B. Critically, directions do not transfer: applying the Qwen3-8B direction to Llama-3-8B yields AUROC 0.571 (near random), and the cross-model direction cosine is only 0.23. This indicates that sycophancy directions are model-specific artifacts rather than universal encodings of sycophantic cognition.

Three-way decomposition. We introduce a neutral condition (neither sycophantic nor genuine revision) to compute an independent genuine-revision direction d_{gen} (60 neutral pairs).

Table 3: Cross-model direction analysis between Qwen3-8B and Llama-3-8B.

Metric	Qwen3-8B	Llama-3-8B
AUROC (own direction, expanded)	0.753	0.712
AUROC (own direction, weak only)	0.811	0.774
Optimal layer	L24	L18
AUROC (Qwen dir. on Llama)	–	0.571
AUROC (Llama dir. on Qwen)	0.563	–
Cross-model cosine	0.23	

Table 4 reports the results. The cosine between d_{gen} and d_{syc} is -0.82, meaning they are near-antiparallel. This implies that sycophancy and genuine revision are not linearly separable independent dimensions; they occupy approximately opposite ends of a single axis. The independent d_{gen} achieves only 0.691 AUROC for genuine-vs-neutral classification, and the three-way R_2 reward provides only 0.008 improvement over the rank-based design.

Table 4: Three-way direction analysis with independent neutral condition.

Metric	Value
d_{gen} vs. d_{syc} cosine	-0.82
d_{gen} AUROC (genuine vs. neutral)	0.691
d_{syc} AUROC (sycophantic vs. neutral)	0.723
d_{syc} AUROC (sycophantic vs. genuine, original)	0.753
Three-way R_2 improvement over rank-based R_2	0.008

4.2 DC-GRPO Training Dynamics

Table 5 reports the trajectory of DC-GRPO training. The key observation is the progressive degradation of R_1: its mean reward drops from 0.55 (step 500) to 0.31 (step 6000), and its correlation with R_3 falls from 0.67 to 0.29. By the second half of training, R_1 contributes noise rather than signal. The model's final behavior is primarily shaped by R_3, with R_1 and R_2 acting as friction.

Table 5: DC-GRPO training dynamics. R_1-R_3 correlation degrades steadily, indicating that the representation-level reward decouples from the behavioral reward during training.

Step	R_1	R_2	R_3	Total	R_1-R_3 Corr.	Cos. to Init	Held-out AUROC
500	0.55	0.48	0.39	0.46	0.67	0.91	0.738
1000	0.52	0.46	0.43	0.46	0.64	0.84	0.721
1500	0.47	0.44	0.46	0.45	0.58	0.76	0.706
2000	0.43	0.41	0.48	0.44	0.52	0.68	0.689
2500	0.40	0.39	0.49	0.43	0.47	0.71	0.694
3000	0.38	0.38	0.50	0.43	0.43	0.64	0.678
4000	0.35	0.36	0.51	0.42	0.37	0.58	0.661
5000	0.33	0.35	0.51	0.41	0.33	0.62	0.653
6000	0.31	0.34	0.52	0.41	0.29	0.55	0.642

For comparison, Table 6 shows the R3-only baseline, which exhibits stable, monotonically increasing reward throughout training.

Table 6: R3-only GRPO training dynamics. Reward increases monotonically without the instability observed in DC-GRPO.

Step	R_3	Total Reward	Training Loss	KL Div.
1000	0.42	0.42	1.51	0.11
2000	0.47	0.47	1.46	0.18
3000	0.50	0.50	1.43	0.23
4000	0.52	0.52	1.41	0.27
5000	0.53	0.53	1.39	0.29
6000	0.54	0.54	1.38	0.31

4.3 Direction Stability During Training

Table 7 tracks d_{syc} stability throughout DC-GRPO training. The cosine similarity between the current and initial direction decreases from 0.91 (step 500) to a minimum of 0.53 (step 5500). The cosine dropped below the 0.7 recalibration threshold 7 times, triggering 4 recalibration events (at steps 2000, 3000, 4000, and 5500). Each recalibration recovers an average of 0.08 cosine, producing a sawtooth pattern that nonetheless trends downward. The held-out AUROC degrades in parallel, from 0.753 (initialization) to 0.642 (step 6000), a total loss of 0.111. The drift rate is approximately 0.062 cosine per 1000 steps.

Table 7: Direction stability timeline during DC-GRPO training. Recalibrations provide temporary recovery but cannot reverse the downward trend.

Step	Cos. to Init	Cos. to Prev	Held-out AUROC	Recalibrated?
0	1.00	–	0.753	Init
500	0.91	0.91	0.738	No
1000	0.84	0.943	0.721	No
1500	0.76	0.928	0.706	No
2000	0.68	0.911	0.689	Yes
2500	0.71	0.936	0.694	No
3000	0.64	0.918	0.678	Yes
3500	0.59	0.924	0.669	No
4000	0.58	0.947	0.661	Yes
4500	0.62	0.938	0.656	No
5000	0.56	0.921	0.653	No
5500	0.53	0.958	0.646	Yes
6000	0.55	0.961	0.642	No

This instability creates a fundamental problem for representation-level rewards: the direction used to compute rewards at step t may not accurately describe the model's activation geometry at step t, turning the reward signal into noise. Even if the initial direction were perfectly accurate, training-induced representation drift would erode its validity.

4.4 R1-R3 Disagreement Analysis

If representation-level rewards capture information that behavioral rewards miss, we should observe cases where R_1 detects sycophancy that R_3 approves (masked sycophancy). Table 8 reports disagreement rates across training phases.

The overall disagreement rate of 3.8% falls below a meaningful threshold. Disagreement is highest for weak-evidence scenarios (5.1%) and lowest for strong evidence (2.8%), as shown in Table 9, consistent with the evidence-strength confound identified in Section 4.1.

Manual review of 50 sampled R_1-sycophantic/R_3-correct cases (Table 10) reveals that only 16% represent clear masked sycophancy. The majority are ambiguous (46%) or R_1 false positives

Table 8: R1-R3 disagreement frequency by training phase. Overall disagreement is 3.8%, below the 5% threshold needed to support a claim of meaningful representation-level advantage.

Phase	N	Disagree	%	R_1-syc/R_3-ok	R_1-ok/R_3-syc
Step 0–2000	6200	285	4.6	168	117
Step 2000–4000	6100	226	3.7	134	92
Step 4000–6000	6100	189	3.1	110	80
Overall	18400	700	3.8	412	289

Table 9: R1-R3 disagreement by evidence strength.

Evidence	N	Disagree %	R_1-syc/R_3-ok %	R_1-ok/R_3-syc %
Weak	7400	5.1	2.9	2.2
Medium	5600	3.4	2.1	1.3
Strong	5400	2.8	1.6	1.2

(38%), further undermining the case for representation-level advantage.

Table 10: Qualitative review of R1-sycophantic/R3-correct disagreement cases ($n = 50$).

Category	Count	%
Clear masked sycophancy	8	16
Ambiguous	23	46
R_1 false positive	19	38

4.5 R1-Only Ablation

To isolate the effect of the representation-level reward, we train an R1-only variant (no R_2 or R_3) for 2000 steps from a step-2000 DC-GRPO checkpoint. Table 11 reports the results.

The R1-only model achieves a high R_1 reward (0.72) but the worst behavioral outcomes: NoF 0.361, warranted revision 0.463, MMLU regression of -3.7 percentage points. We identified 34 reward hacking instances where the model generates lengthy hedging prefixes that project away from d_{syc} in activation space without actually reducing sycophantic behavior. This confirms that R_1 reward can be optimized without improving the target behavior, a systematic risk of representation-level rewards that lack behavioral grounding.

4.6 Turn Depth Analysis

Table 12 and Table 13 report sycophancy metrics stratified by conversation turn depth (turn 2, 3, and 4+).

All trained methods reduce the commitment-depth gradient relative to vanilla (from 0.213 to approximately 0.14), but DC-GRPO shows no differential advantage at any depth. The gap between DC-GRPO and R3-only is consistent across turns (1.5 to 2.1 percentage points on NoF), indicating that the directional signal does not capture commitment-depth-specific sycophancy patterns beyond what behavioral rewards detect.

4.7 Full Evaluation

Table 14 presents the complete evaluation across all methods and metrics. All trained methods significantly reduce unwarranted capitulation relative to vanilla ($p < 0.001$), but no trained method significantly outperforms any other ($p > 0.34$ for all pairwise comparisons, Table 15).

Table 11: R1-only ablation compared to DC-GRPO, R3-only, and vanilla baselines.

Metric	R1-only	DC-GRPO	R3-only	Vanilla
NoF ↓	0.361	0.318	0.302	0.412
WR ↑	0.463	0.527	0.538	0.583
MMLU	0.584	0.598	0.609	0.621
MT-Bench	6.51	6.84	7.02	7.18
R_1 reward	0.72	0.31	–	–

Table 12: NoF by turn depth. All methods show similar depth gradients; DC-GRPO does not outperform baselines at any depth.

Model	Turn 2	Turn 3	Turn 4+	Gradient
Vanilla	0.318	0.428	0.531	0.213
R3-only	0.241	0.316	0.381	0.140
ACT	0.253	0.322	0.393	0.140
Binary Probe	0.247	0.319	0.388	0.141
DC-GRPO	0.258	0.331	0.402	0.144

DC-GRPO achieves the worst NoF (0.318), worst MMLU (0.598), and worst MT-Bench (6.84) among trained methods. The binary probe baseline, which is strictly simpler, matches or exceeds DC-GRPO on every metric. The ACT baseline with its undifferentiated consistency loss is also statistically indistinguishable from DC-GRPO.

The double-bind persists. Table 16 shows the revision-capitulation trade-off. All methods reduce unwarranted capitulation (from 0.378 to 0.268–0.282) at the cost of warranted revision (from 0.583 to 0.498–0.538). No method resolves this trade-off, and DC-GRPO does not improve it relative to baselines.

4.8 Binary Probe Comparison

Table 17 provides a direct comparison between the binary probe baseline and DC-GRPO. The probe suffers from the same direction-drift phenomenon (AUROC degrades from 0.761 to 0.651 over training), but its impact is more contained because a scalar probability output is more robust to representation shifts than a directional projection. No metric difference is statistically significant.

This result challenges the necessity of directional decomposition. A simpler binary classifier achieves equivalent performance with fewer moving parts (no contrastive R_2, no multi-component reward weighting, no directional recalibration).

5 Discussion

Why does detection fail to transfer? Our experiments identify a cascade of failures connecting detection to training. First, the direction itself encodes confounding features: the AUROC gradient from 0.811 (weak evidence) to 0.698 (strong evidence) means that 30–40% of the signal reflects evidence-strength context rather than sycophantic cognition. Second, even the genuine signal component degrades during training: LoRA updates shift the representation space faster than recalibration can track, and direction AUROC drops from 0.753 to 0.642 over 6000 steps. Third, the three-way decomposition reveals that d_{syc} and d_{gen} are near-antiparallel (cosine $= -0.82$), meaning that linear probing in this domain captures a contrast axis ("being persuaded" vs. "not being persuaded") rather than two independent constructs. Together, these factors make R_1 a noisy, drifting signal that contributes friction rather than guidance.

Table 13: Warranted revision rate by turn depth.

Model	Turn 2	Turn 3	Turn 4+
Vanilla	0.628	0.576	0.524
R3-only	0.592	0.531	0.472
ACT	0.553	0.491	0.432
Binary Probe	0.586	0.524	0.465
DC-GRPO	0.581	0.518	0.461

Table 14: Full evaluation summary. DC-GRPO ranks last among trained methods on most metrics.

Model	NoF ↓	ToF ↑	WR ↑	UC ↓	RC-AUC ↑	SycEval ↑	MMLU	MT-B
Vanilla	.412	.701	.583	.378	.602	.548	.621	7.18
R3-only	.302	.653	.538	.268	.672	.634	.609	7.02
ACT	.310	.628	.498	.274	.651	.608	.605	6.91
Bin. Probe	.308	.644	.531	.272	.669	.627	.607	6.97
DC-GRPO	.318	.641	.527	.282	.668	.621	.598	6.84

Implications for representation-level reward methods. The direction instability problem generalizes beyond our specific setup. Any method that uses internal model states as reward signals during RL training faces the same challenge: the states that define the reward change as the policy updates. CRAFT [Zhang et al., 2024] addresses this through prototype clustering with periodic updates, but our results suggest that even frequent recalibration (every 500 steps) is insufficient when the underlying direction is confounded. Linear Probe Penalties [Wang et al., 2024] may be more robust by using the reward model's activations rather than the policy's, but this comes at the cost of detecting policy-specific internal states.

The double-bind as a shared limitation. All methods we tested reduce unwarranted capitulation at the cost of warranted revision. The warranted revision rate drops from 0.583 (vanilla) to 0.498–0.538 across trained models. This is not a failure of any particular method but a reflection of the difficulty of the underlying problem: at the 8B parameter scale, the internal distinction between "I am being persuaded by good evidence" and "I am caving to social pressure" may not be linearly separable in activation space. Future work might investigate non-linear probing methods or larger model scales where the distinction could be more salient.

Limitations. Our experiments use a single primary model (Qwen3-8B) at a single scale (8B parameters). The negative results may not hold at larger scales where sycophancy directions could be more discriminative. The calibration data, while style-controlled, covers a limited range of dialogue scenarios. We also use LoRA rather than full fine-tuning, which constrains the representation space in specific ways.

6 Conclusion

We set out to test whether activation-space sycophancy detection can be leveraged as a reward signal for GRPO training. The answer, at 8B scale with our setup, is no. Direction-Contrastive GRPO provides no measurable benefit over a behavioral-reward-only baseline while incurring greater capability degradation. We trace this failure to three structural causes: evidence-strength confounds in the sycophancy direction, direction instability under training, and the near-antiparallel geometry of sycophantic and genuine revision directions. These findings do not invalidate activation-space analysis of sycophancy as a measurement tool, but they establish

Table 15: Pairwise bootstrap significance tests on NoF. No pair of trained methods differs significantly.

Comparison	NoF Diff	p-value	Sig. at 0.05?
DC-GRPO vs. R3-only	0.016	0.35	No
DC-GRPO vs. ACT	0.008	0.62	No
DC-GRPO vs. Bin. Probe	0.010	0.54	No
R3-only vs. ACT	−0.008	0.61	No
R3-only vs. Bin. Probe	−0.006	0.68	No
ACT vs. Bin. Probe	0.002	0.91	No
Any trained vs. Vanilla	∼−0.10	<0.001	Yes

Table 16: Revision-capitulation trade-off. No method resolves the double-bind between reducing sycophancy and preserving genuine revision.

Model	WR ↑	UC ↓	RC-AUC ↑	Double-bind resolved?
Vanilla	0.583	0.378	0.602	No (high both)
R3-only	0.538	0.268	0.672	Partial
ACT	0.498	0.274	0.651	No (WR drop > UC drop)
Bin. Probe	0.531	0.272	0.669	Partial
DC-GRPO	0.527	0.282	0.668	No

that detection accuracy in a static setting does not predict utility as a dynamic training signal. We release our expanded calibration dataset and direction stability monitoring code to support further investigation.

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Table 17: Binary Probe vs. DC-GRPO head-to-head comparison.

Metric	Bin. Probe	DC-GRPO	Diff	p
NoF	0.308	0.318	0.010	0.54
WR	0.531	0.527	−0.004	0.82
RC-AUC	0.669	0.668	−0.001	0.94
MMLU	0.607	0.598	−0.009	0.38
Final probe/dir. AUROC	0.651	0.642	−0.009	–

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